

CS 2B 01357101 黃翊宏 (1人)

Computer Networking HW2

1. List five nonproprietary Internet applications and the application-layer protocols that they use.

- | | | |
|-----------------------|---|----------|
| 1. file transfer | — | FTP |
| 2. e-mail | — | SMTP |
| 3. Web-documents | — | HTTP 1.1 |
| 4. Internet telephony | — | SIP, RTP |
| 5. Network Time | — | NTP |

2. Caching, for example DNS RR caching or Web page caching, is one of useful techniques in computer systems. Describe what are possible benefits to use caching mechanism. And, at the same time, what are possible side-effects (in the bad sides) we much be aware of when caching mechanisms are applied?

優點：減少回應時間、降低網路頻寬消耗、減輕原始伺服器負載、提高服務可用性。

缺點：資源過時、額外儲存開銷和資安風險。

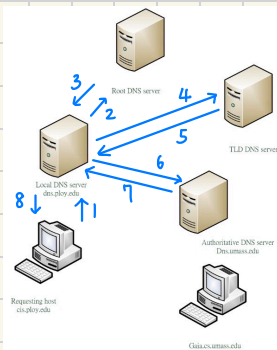
3. DNS. (Application layer)

In general, there are two mechanisms for DNS domain name resolving (that is, find out the IP of a given domain name), iterative and recursive. Refer to Figure 1, suppose now the host “cis.poly.edu” would like to know the IP addresses for the domain name “gaia.cs.umass.edu”.

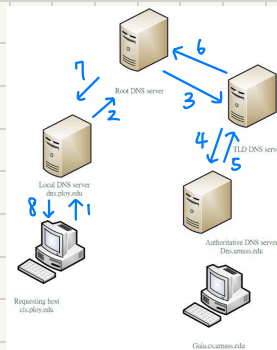
- Please draw DNS query message flow (both for dns request and reply) for the two mechanisms individually. (Assume no cache results in any of the hosts or servers.)
- How DNS servers know the IP addresses for root DNS servers?
- Why not centralize DNS system?
- By default, DNS servers use iterative mechanism for name resolving. What are possible reasons?
- For each request (both in iterative/recursive), we might need to query root DNS server. The number of requests should be huge. Why we, in practices, just need few number of root servers (e.g. 14-16 worldwide) to handle the requests.
- How DNS cache works? Why use DNS cache?

(a)

iterative :



recursive :



(b)

所有的DNS伺服器都內建了一個“Root Hints”的設定檔，內含全球13組根伺服器的位址，因此所有DNS Server都會知道怎麼連絡Root DNS。

(c)

1. 避免單點故障
2. 流量過大無法負荷
3. 距離過遠產生高延遲
4. 難以維護

(d)

recursive query 會將解析的工作放在被詢問的server上，這會導致越上層的DNS Server 負載越高。Iterative query 則將工作量分散回各local DNS。

(e)

因為local name servers 通常會快取TLD Server 的位址，實務上很少真的去訪問Root name server.

1 2
(f). How DNS cache works? Why use DNS cache?

1. 當一 name server 得到新的對應關係後, 將其暫時保存起來, 並在 TTL 到期後自動刪除。
2. 減少回應時間、降低網路頻寬消耗

4. DNS VULNERABILITIES.

We have seen that DNS is a critical component of the Internet infrastructure, and so it is quite nature to be an attacking target. How can DNS be attacked? For example, you can find different types of DNS Attacks from internet (e.g. <https://www.paloaltonetworks.com/cyberpedia/what-is-a-dns-attack>.) Please give 3 possible DNS attach approaches. For each of them, please give a short description about how it works.

1. DDoS attacks: 發送大量流量給 DNS Server 使其無法處理正常請求。
2. Redirect attacks: 攔截或偽造 DNS 回應來將使用者導向至錯誤位址。
3. Exploit DNS for DDoS: 攻擊者發送 DNS 查詢, 但將來源位址偽裝成受害者 IP。

5. Streaming multimedia: DASH (Dynamic, Adaptive Streaming over HTTP)

- (a). What's main reason to send streaming data over HTTP ?
- (b). Why DASH chops the video file into small pieces (chunks)?
- (c). How DASH deal with heterogeneous clients (that is, bandwidth available for different users might not be the same, some are with rich bandwidth some are not) ?
- (d). How DASH adapts to highly dynamics of internet traffics (for example, end-to-end bandwidth and packet delays are highly unpredictable)?

(a) 利用 HTTP 協定來傳送串流資料可以很好的複用現有的 Web 基礎設施, 並使用 Web Cache 和 CDN 進行大規模分發。

(b) 方便動態調整調整畫質, 減少緩衝壓力。

(c) Server 會將相同片段、但品質不同的檔案都先儲存起來, 並提供一個 manifest file 給 Client 讓其自主選擇。

(d) client 會週期性的測量從 server 到 client 的之間頻寬, 並根據測量結果與緩衝區狀態決定什麼時候請求; 以及請求片段的品質。

6. Content Distribution Networks (CDNs)

- (a) what are the main services provided by CDN
- (b) what are possible benefits to use CDN for a content provider, eg. Netflix
- (c) CDN can effectively mitigate DDoS attacks, why?

- (a) 資源快取與分散式儲存內容、負載平衡、動態資源請求
- (b) 避免單點故障、降低網路擁塞、減少延遲、提高服務擴展性。
- (c) 因為CDN分散式的特性，攻擊者很難同時癱患所有節點。且由於CDN消除了單點故障的可能性，即使單一節點癱患，流量請求仍可分散到其他臨近節點。

7. OTT media service vs. IPTV

- (a) currently, media services can be provided via OTT or IPTV, what are the difference between them (you may want to refer to <https://otter-video.com/blog/iptv-vs-ott/>)
- (b) give an example of OTT media service provider in Taiwan (or your country), and an example for IPTV provider.
- (c) In your opinion, which one (OTT or IPTV) might win in media service market? why?

(a)	IPTV	OTT
網路環境	電信商提供的專用網路	現有的公共網際網路
服務品質	專用頻寬、能提供幾乎不卡頓的高畫質影像。	畫質與體驗取決於使用者當下網路環境
觀看設備	需要機上盒進行訊號解碼	任何能存取公共網際網路的設備
傳輸技術	多採用 Multicast	Unicast 配合 DASH + CDN

- (b) OTT Provider: 巴哈姆特動畫瘋
IPTV Provider: 中華電信 MOD

- (c) 依現行市場情況來看，OTT應已勝出且具主導地位。
現代人時間零碎，已無法在電視前久待，而OTT跨裝置的特性正是其優勢，再加上原創內容的豐富度、訂閱取消流程更簡單、不需額外機上盒，讓OTT已成為主流。

8. Internet tools: nslookup, traceroute, whois

- (a) Please use **nslookup** to check what is the IP address (or addresses) of www.facebook.com and www.google.com. Write down what you found and compare your answer with your classmates and check if there are any differences.
- (b) Please use **traceroute (or tracert)** command to check the routing path to the server
- (c) Please use **whois** command to check the owner of the IP address.

(a)

```
nslookup www.facebook.com
Server: 192.168.50.1
Address: 192.168.50.1

Non-authoritative answer:
Name: www.facebook.com
Address: 157.140.2.254
Address: 157.140.2.254
Address: 157.140.2.254
Address: 157.140.2.254

nslookup www.google.com
Server: 192.168.50.1
Address: 192.168.50.1

Non-authoritative answer:
Name: www.google.com
Address: 142.251.153.119
Address: 142.251.153.119
Address: 142.251.153.119
Address: 142.251.153.119

nslookup www.facebook.com
Server: 192.168.50.1
Address: 192.168.50.1

Non-authoritative answer:
Name: www.facebook.com
Address: 157.140.2.254
Address: 157.140.2.254
Address: 157.140.2.254
Address: 157.140.2.254

nslookup www.google.com
Server: 192.168.50.1
Address: 192.168.50.1

Non-authoritative answer:
Name: www.google.com
Address: 142.251.153.119
Address: 142.251.153.119
Address: 142.251.153.119
Address: 142.251.153.119
```

經比較，nslookup 的結果可能因使用者所處位置、DNS Server 和時間有關。

這可能是因不同 DNS Server 所持有的快取不同，又或者是為了負載平衡，將使用者動態分配到其他副本伺服器上。

(b)

```
tracert www.google.com
Tracing route to google.com [142.251.153.119] over 30 hops:
 0  0 ms  0 ms  0 ms  RT-AXS6U-V2-BAC0 [192.168.50.1]
 1  <1 ms  <1 ms  <1 ms  RT-AXS6U-V2-BAC0 [192.168.50.1]
 2  2 ms  2 ms  2 ms  h254.s98.ts.hinet.net [168.95.90.254]
 3  2 ms  2 ms  2 ms  168-95-04-26.tp4-3333.hinet.net [168.95.04.26]
 4  3 ms  3 ms  3 ms  228-128-0-262.tyfo-3033.hinet.net [228.128.0.262]
 5  3 ms  3 ms  3 ms  228-128-0-13.tyfo-3335.hinet.net [228.128.0.13]
 6  4 ms  4 ms  4 ms  72.14.215.8
 7  4 ms  4 ms  3 ms  142.251.153.119

tracert www.facebook.com
Tracing route to star-mini.c10r.facebook.com [31.13.07.36] over 30 hops:
 0  1 ms  1 ms  1 ms  RT-AXS6U-V2-BAC0 [192.168.50.1]
 1  2 ms  2 ms  2 ms  h254.s98.ts.hinet.net [168.95.90.254]
 2  3 ms  3 ms  3 ms  168-95-02-26.tp4-3332.hinet.net [168.95.02.26]
 3  4 ms  4 ms  4 ms  228-128-14-93.pcpd-3332.hinet.net [228.128.14.93]
 4  2 ms  2 ms  2 ms  218-05-126-237.hinet-jp.hinet.net [218.05.126.237]
 5  3 ms  3 ms  3 ms  ps002.sme41.tpel1.tfbw.net [129.134.49.20]
 6  2 ms  2 ms  2 ms  psw01.tpel1.tfbw.net [173.252.64.108]
 7  6 ms  6 ms  5 ms  mswlap.01.tpel1.tfbw.net [129.134.92.178]
 8  2 ms  2 ms  2 ms  edge-star-mini-shv01-tpel1.facebook.com [31.13.07.36]
```

(c)

facebook.com

Domain Information

Domain: facebook.com

Registered On: 1997-03-29

Expires On: 2034-03-30

Updated On: 2025-04-23

Status: client delete prohibited
client transfer prohibited
client update prohibited
server delete prohibited
server transfer prohibited
server update prohibited

Name Servers: a.ns.facebook.com
b.ns.facebook.com
c.ns.facebook.com
d.ns.facebook.com

Registrar Information

Registrar: RegistrarSafe, LLC

IANA ID: 3237

Abuse Email: abusecomplaints@registrar-safe.com

Abuse Phone: +1-650-308-7004

google.com

Domain Information

Domain: google.com

Registered On: 1997-09-15

Expires On: 2029-09-14

Updated On: 2019-09-09

Status: client delete prohibited
client transfer prohibited
client update prohibited
server delete prohibited
server transfer prohibited
server update prohibited

Name Servers: ns1.google.com
ns2.google.com
ns3.google.com
ns4.google.com

Registrar Information

Registrar: MarkMonitor Inc.

IANA ID: 292

Email: https://www.markmonitor.com/contact-us/

Abuse Email: abusecomplaints@markmonitor.com

Abuse Phone: +1.208851750

Registrant Contact

Organization: Google LLC

Country: US

Email: https://domains.markmonitor.com/whois/contact/google.com

Technical Contact

Email: https://domains.markmonitor.com/whois/contact/google.com